

Capstone Progress Documentation – Week 9

Project ID: PW26_MG_02

Objective

Week 9 involved balancing ISA-1 exams along with steady security development. During system reviews, we found JSON in critical components causing delay and ambiguity. We worked to replace these areas with stricter binary mechanisms. This cleanup is helping us prepare the system for secure multimodal Explainable AI (XAI) in the next phase.

Progress Update

GhostKey & InsiderQuantum Drift Defense

GhostKey refers to hidden credential misuse, and InsiderQuantum Drift means small internal model changes that slowly affect results. During the Cloud Computing ISA phase, we reviewed key transfer and parameter updates. We replaced flexible JSON handling with shared-memory binary sync and strict schemas to improve reliability. Clean and structured data flow will support accurate multimodal XAI tracing later.

ModelDrift & Regulatory Defense

ModelDrift means corruption in model communication, and Regulatory Defense means preventing small compliance issues from growing. Around the OOAD ISA, we improved telemetry and compliance logging. Deterministic binary validation replaced unclear JSON logs. Clear monitoring across layers now supports transparent multimodal reasoning for XAI.

ShadowSupply & StreamHijack Defense

ShadowSupply highlights hidden risks in third-party libraries, and StreamHijack refers to tampering in live market feeds. During the Compiler Design ISA period, we secured dependency files and data streams using signed binaries and packet validation. Trusted inputs are essential for reliable multimodal explainability.

Dark Pool, Latency & Fraud Controls

Dark Pool detection identifies hidden liquidity distortion, Latency control prevents timing exploitation, and Fraud controls detect spoofing and fake trades. In the final ISA phase, we improved liquidity tracking, timing precision, and anomaly detection using binary graphs and buffers. Structured execution data will allow clearer multimodal XAI insights.

Outcome

By the end of the week, key weaknesses were resolved and system clarity improved. The architecture is now more stable and secure, forming a strong base for explainable and trustworthy multimodal intelligence.

Next Steps

Next week, we will build an Explainability Toolkit to trace decisions across models, data, and execution steps. With stronger structure in place, we are ready to move toward secure and transparent multimodal XAI.

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